Problem: A
Board Game

You are playing a board game where a token starts at position 0, and your goal is to reach position n.

On each move, you roll a standard 6-sided die and advance the token forward by the number of steps shown on the die. Thus, in one move, the token can move forward by 1 to 6 positions.

Determine how many different sequences of die rolls will move the token from position 0 to exactly position n.

Constraints:

- $1 \leq n \leq 10^6$
- Since the answer can be large, print the result modulo $10^9 + 7$.

Input:
A single integer n — the target position to reach.

Output:
Print the number of different sequences of dice rolls that sum up to n, modulo $10^9 + 7$.

Sample Input 1:

```
3
```

Sample Output 1:

```
4
```

Sample Input 2:

```
7
```

Sample Output 2:

```
63
```

Sample Input 3:

```
10
```

Sample Output 3:

```
492
```

Problem: B
Book Shop

You are in a book shop which sells n different books. You know the price and number of pages of each book.

You have decided that the total price of your purchases will be at most x. What is the maximum number of pages you can buy? You can buy each book at most once.

Input:

- The first input line contains two integers n and x: the number of books and the maximum total price.
- The next line contains n integers $h_1, h_2, \ldots, h_n$: the price of each book.
- The last line contains n integers $s_1, s_2, \ldots, s_n$: the number of pages of each book.

Output:

- Print one integer: the maximum number of pages.

Constraints:

- $1 \leq n \leq 1000$
- $1 \leq x \leq 10^5$
- $1 \leq h_i, s_i \leq 1000$

Sample Input:

```
4 10
4 8 5 3
5 12 8 1
```

Sample Output:

```
13
```